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<https://github.com/ChanceValdez/APM1110/tree/main/FA3>

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Exercise 6.1

Item 5

A geospatial analysis system has four sensors supplying images. The percentage of images supplied by each sensor and the percentage of images relevant to a query are shown in the following table.

Sensor	Percentage of Images Supplied	Percentage of Relevant Images
1	15	50
2	20	60
3	25	80
4	40	85

What is the overall percentage of relevant images?

Step by Step Computation

According to Theorem 6.1, the Law of Total Probability, if a sample space S can be partitioned into k mutually exclusive and exhaustive events, $A_1, A_2, A_3, \dots, A_k$, then for any event E :

$$P(E) = P(A_1)P(E | A_1) + P(A_2)P(E | A_2) + \dots + P(A_k)P(E | A_k)$$

Let:

- $k = 4$, representing the total number of image sensors.
- S_i be the event that an image is supplied by Sensor i , where $i \in \{1, 2, 3, 4\}$.
- R be the event that a selected image is relevant to the query.
- $P(R | S_i)$ be the conditional probability that an image is relevant, given it is from Sensor i .

The total probability formula for our specific events is:

$$P(R) = P(S_1)P(R | S_1) + P(S_2)P(R | S_2) + P(S_3)P(R | S_3) + P(S_4)P(R | S_4)$$

Substituting our values into the formula:

$$\begin{aligned} P(R) &= P(S_1)P(R | S_1) + P(S_2)P(R | S_2) + P(S_3)P(R | S_3) + P(S_4)P(R | S_4) \\ &= (0.15 \times 0.50) + (0.20 \times 0.60) + (0.25 \times 0.80) + (0.40 \times 0.85) \\ &= 0.075 + 0.120 + 0.200 + 0.340 \\ &= 0.735 \end{aligned}$$

Thus, the overall percentage of relevant images is 73.5%.

R Code for Verification

```
sensor_data <- data.frame(
  Sensor = 1:4,
  Supplied = c(0.15, 0.20, 0.25, 0.40),
  Relevant = c(0.50, 0.60, 0.80, 0.85)
)
sensor_data

##   Sensor Supplied Relevant
## 1      1      0.15    0.50
## 2      2      0.20    0.60
## 3      3      0.25    0.80
## 4      4      0.40    0.85

rel_img <- sum(sensor_data$Supplied * sensor_data$Relevant)

cat("Overall Percentage of Relevant Images:", rel_img * 100, "%.", sep = " ")

## Overall Percentage of Relevant Images: 73.5 %.
```

Item 6

A fair coin is tossed twice.

Let E_1 be the event that both tosses have the same outcome, that is, $E_1 = \{HH, TT\}$.

Let E_2 be the event that the first toss is a head, that is, $E_2 = \{HH, HT\}$.

Let E_3 be the event that the second toss is a head, that is, $E_3 = \{TH, HH\}$.

Show that E_1 , E_2 , and E_3 are pairwise independent but not mutually independent.

Step by Step Computation

Let the sample space S be the set of all possible outcomes when a fair coin is tossed twice:

$$S = \{HH, HT, TH, TT\}$$

As the coin is fair, each of the 4 outcomes is equally likely, meaning the probability of any individual outcome is $\frac{1}{4}$.

$$P(HH) = P(HT) = P(TH) = P(TT) = \frac{1}{4}$$

Using the addition rule for mutually exclusive events, where $P(E \cap F) = 0$, we sum the probabilities of the individual outcomes:

- $P(E_1) = P(HH \cup TT) = P(HH) + P(TT) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$
- $P(E_2) = P(HH \cup HT) = P(HH) + P(HT) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$
- $P(E_3) = P(TH \cup HH) = P(TH) + P(HH) = \frac{1}{4} + \frac{1}{4} = \frac{1}{2}$

Pairwise Independence Test

To show that the events are pairwise independent, we use the independence formula for each pair of events:

$$P(A \cap B) = P(A) \cdot P(B)$$

- E_1 and E_2 :
 $E_1 \cap E_2 = \{HH\} \implies P(E_1 \cap E_2) = \frac{1}{4}$
 $P(E_1) \cdot P(E_2) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$
Since $\frac{1}{4} = \frac{1}{4}$, they are independent.
- E_1 and E_3 :
 $E_1 \cap E_3 = \{HH\} \implies P(E_1 \cap E_3) = \frac{1}{4}$
 $P(E_1) \cdot P(E_3) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$
Since $\frac{1}{4} = \frac{1}{4}$, they are independent.
- E_2 and E_3 :
 $E_2 \cap E_3 = \{HH\} \implies P(E_2 \cap E_3) = \frac{1}{4}$
 $P(E_2) \cdot P(E_3) = \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{4}$
Since $\frac{1}{4} = \frac{1}{4}$, they are independent.

As the formula holds true for all pairs, E_1 , E_2 , and E_3 are **pairwise independent**.

Mutual Independence Test

To show that the events are mutually independent, they must also satisfy the joint formula for all three events together:

$$P(E_1 \cap E_2 \cap E_3) = P(E_1) \cdot P(E_2) \cdot P(E_3)$$

- $E_1 \cap E_2 \cap E_3 = \{HH\} \implies P(E_1 \cap E_2 \cap E_3) = \frac{1}{4}$
- $P(E_1) \cdot P(E_2) \cdot P(E_3) = \frac{1}{2} \cdot \frac{1}{2} \cdot \frac{1}{2} = \frac{1}{8}$

As $\frac{1}{4} \neq \frac{1}{8}$, the formula does not hold. Thus, the events are **not mutually independent**.

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